Paper Review • Vision-Language-Action

HiFOR

Hierarchical Foresight with Action-Guided World Grounding for Vision-Language-Action Model

Letting VLAs foresee the next step, milestone and goal,**then retrieve what the current action needs**

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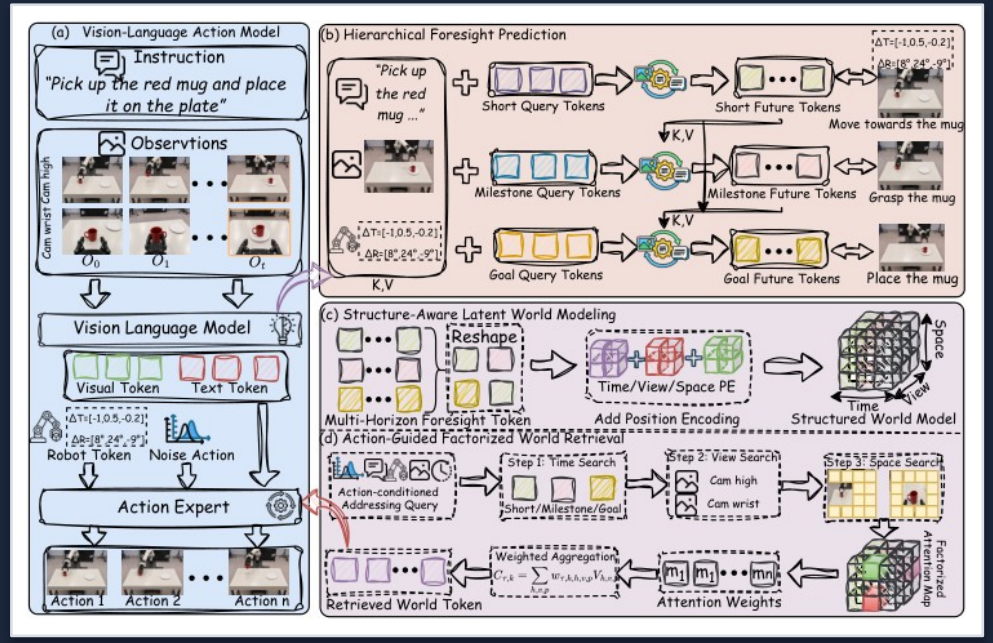


Figure: HiFOR overall framework (paper Fig. 2)

In one sentence: multi-horizon foresight + on-demand retrieval yields consistent gains over π0.5

HiFOR jointly predicts, in DINOv2 feature space, **short-term states** / **the next semantic milestone** / **the final goal**, chained by a hierarchical dependency mask; predictions form a **horizon × view × space** structured latent world, from which the action expert, at **every denoising step**, retrieves on demand in a factorized way to guide action generation.

49.40%

RoboTwin 2.0 Hard

π0.5 † 40.35% → +9.05

98.3%

LIBERO 4-suite avg.

π0.5 96.9% → +1.4

+3.4

LIBERO-Long gain (pts)

92.4% → 95.8%

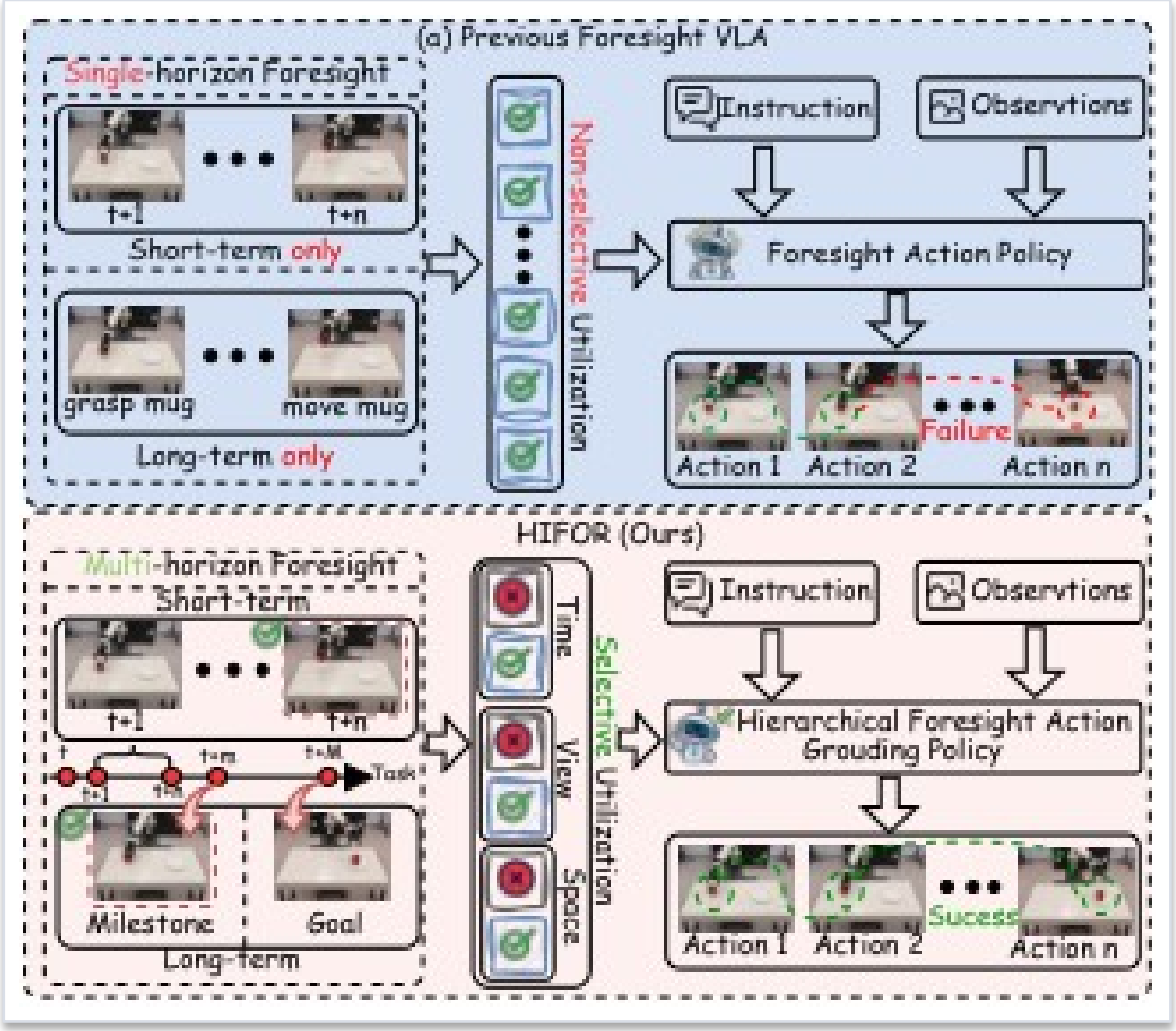
81.7%

Real-robot 6-task avg.

π0.5 70.7% → +11.0

† π0.5 results reproduced by the authors; the real-robot baseline of 70.7% is the average of the six task values in Fig. 4.

Two gaps in existing foresight VLAs: a single time scale, and predictions fed to the policy indiscriminately



Gap ① Fragmented time scales

Short-term prediction captures only local motion; long-term prediction (subtasks / subgoal images) lacks continuous dynamics. Modeled independently, they lack an explicit local evolution → milestone → goal dependency.

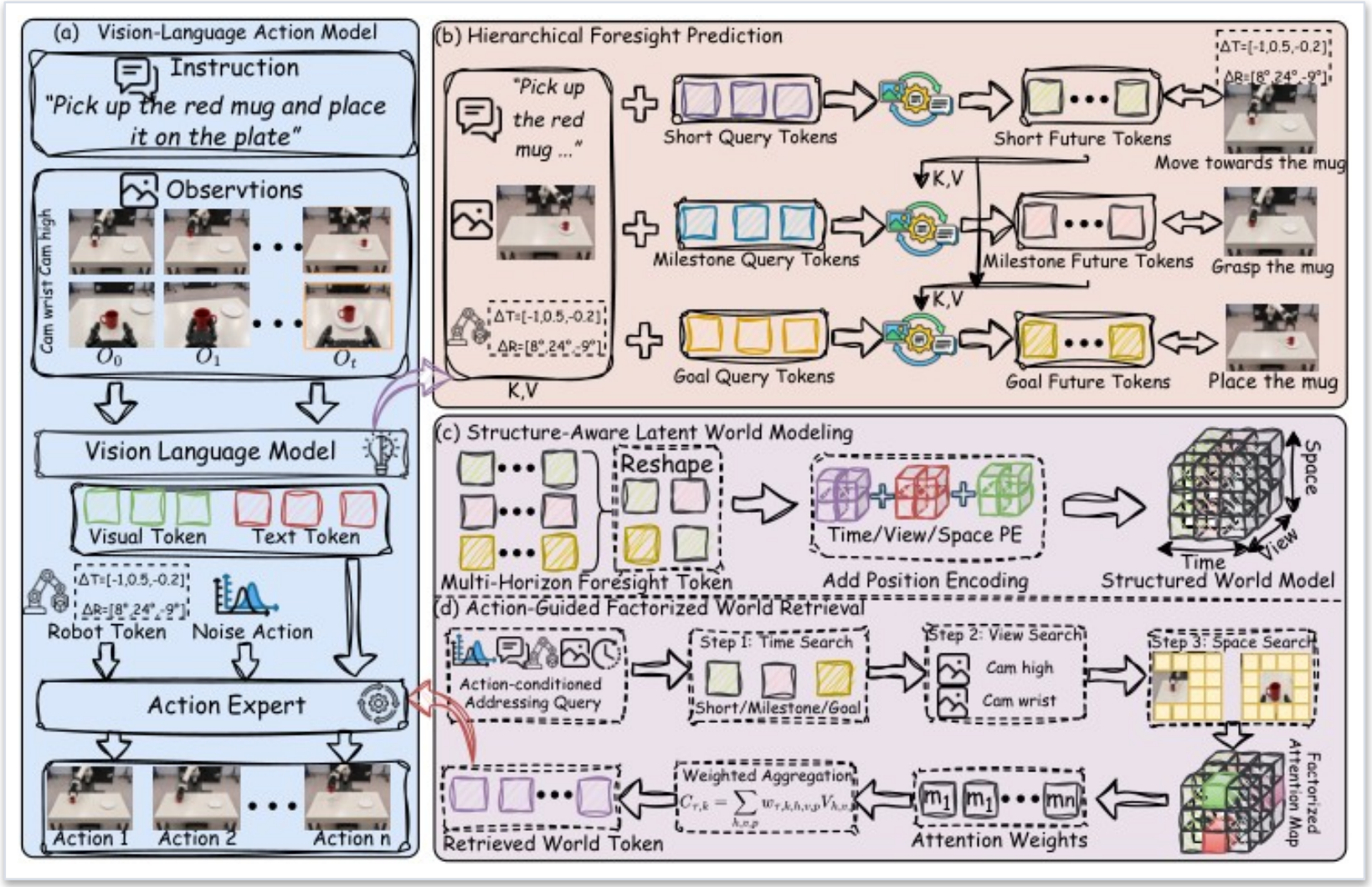
→ HiFOR: hierarchical foresight prediction + HDM

Gap ② Future info used wholesale

Predicted future images / latent features are fed directly as conditions, yet relevance differs across horizons, views and spatial regions, and shifts with execution stage and denoising.

→ HiFOR: structured latent world + action-guided retrieval

Framework: a "predict → organize → retrieve" add-on atop the VLA backbone



① Hierarchical Foresight

Short / Milestone / Goal query sets + HDM, regressing DINOv2 features (plus states for short-term)

② Structured Latent World

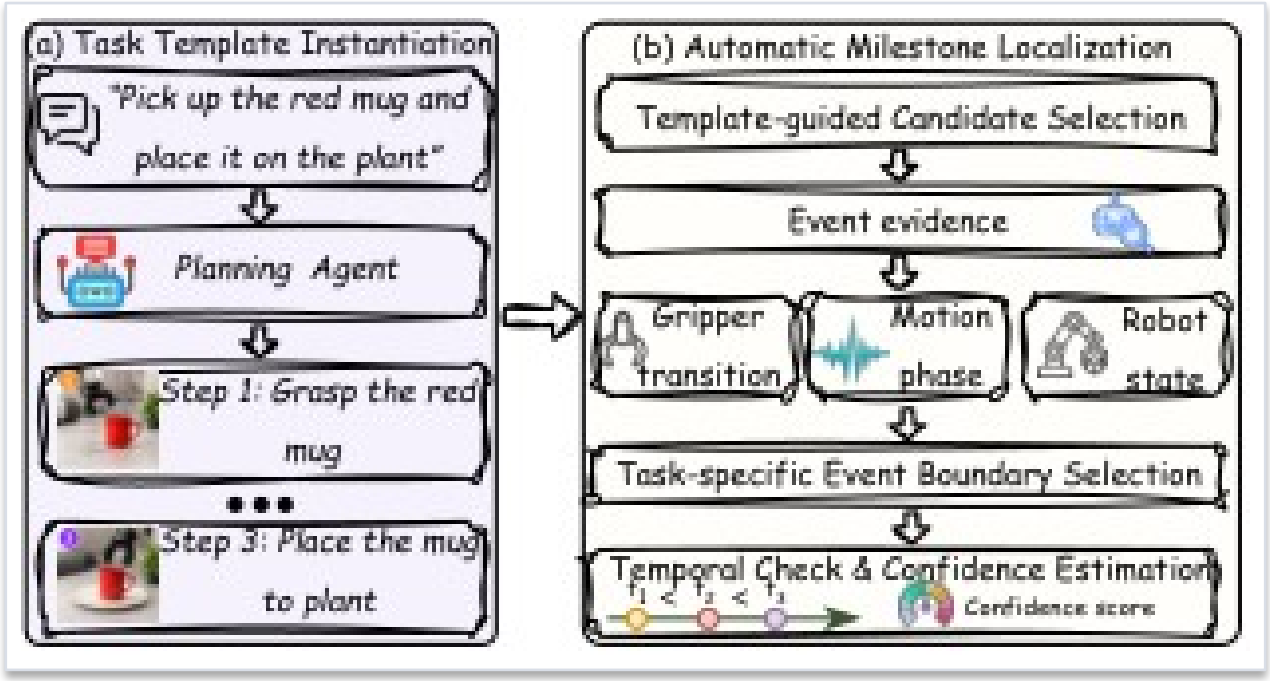
Add temporal / view / spatial positional encodings to get W ∈ ℝ^{3×V×16×d}

③ Action-Guided Retrieval

Action latent at denoising step τ as query: temporal → view → spatial three-level attention

L = L_ACT + L_HFP

Supervision: semantic milestones auto-labeled from existing successful demos, no new trajectories collected



Example: "Pick up the red mug and place it on the plate" → grasp mug / move / place

1 **Task Template**

Planning Agent decomposes the instruction into ordered subgoals $G_1 \ldots G_K$ (stage / event type / completion criterion)

2 **Candidate Boundaries**

Gripper open/close, motion phases, robot state → candidate event boundaries

3 **Event Localization**

Steady-state events: earliest frame where the condition persists; dynamic events: motion / state transition points

4 **Consistency Check**

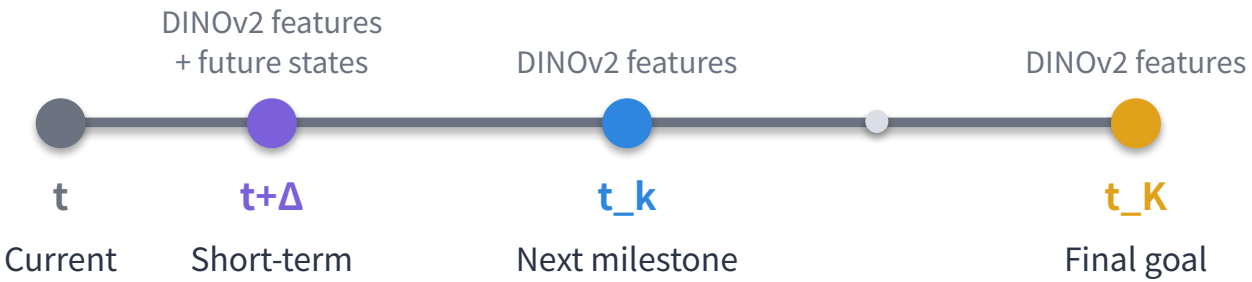
$t_1 < t_2 < \ldots < t_K$; confidence scoring, low-confidence samples manually reviewed

Visual target: frozen DINOv2 ViT-L/14, no prefix, 4×4 adaptive pool

Per view $F \in \mathbb{R}^{16 \times 1024}$, target for 3 horizons

Hierarchical foresight: three horizons share one predictor; HDM passes dependencies one-way, level by level

Three prediction targets (at each time t)



$t^s = \min(t+\Delta, t_K)$ $t^m = \min_{k: t_k > t} t_k$ $t^g = t_K$

Why add state supervision to short-term?

Visual features at the target time alone cannot constrain fine-grained robot motion, so the short-term branch also predicts $s_{t+1}\ldots s_{t^s}$.

Loss: $L_{HFP} = \lambda \cdot D_{vis}(\hat{Z}, Z) + \lambda' \cdot D_{state}(\hat{S}, S)$

HDM: row=query, col=visible token

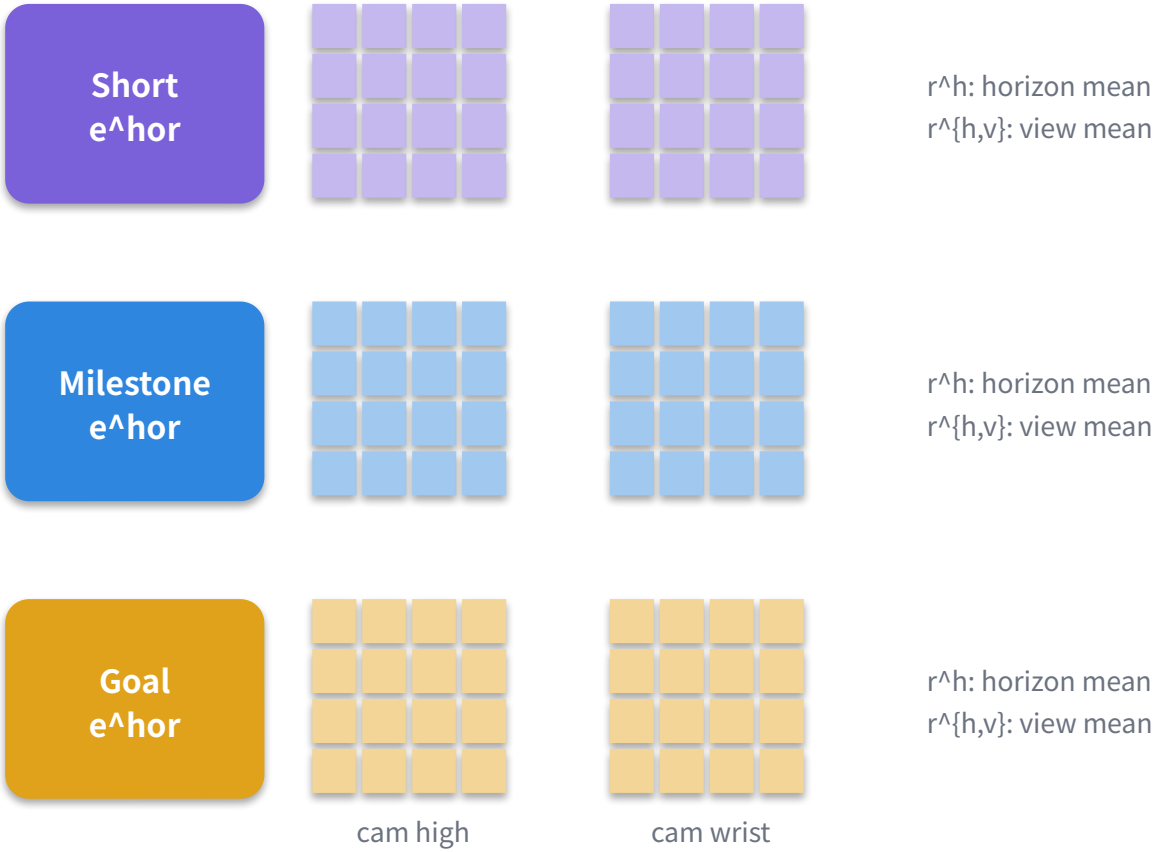
	VLM token Vision + language	Q^s	Q^m	Q^g
Q^s short-term	✓	✓	—	—
Q^m milestone	✓	✓	✓	—
Q^g goal	✓	✓	✓	✓

$U^h = H_\theta(Q^h ; [V_t ; E_L ; Q^{\leqslant h}])$

Three sets of learnable queries share one predictor H_θ; visibility widens along s → m → g, forming a one-way dependency from local dynamics to intermediate progress to the final state.

Structured world + factorized retrieval: actions address future info via "time → view → space"

Structured latent world $W_t \in \mathbb{R}^{3 \times V \times 16 \times d_w}$



$w = P_w(\hat{Z}[h,v,p]) + e^{hor}_h + e^{view}_v + e^{spa}_p$ (each cell = 1 spatial token)

AWR: action latent H^a at denoising step τ as query

Step 1 Temporal — α^h = Softmax over {s, m, g}
key: horizon summary r^h

Step 2 View — $\beta^{v|h}$ = Softmax over V views
key: view summary $r^{h,v}$

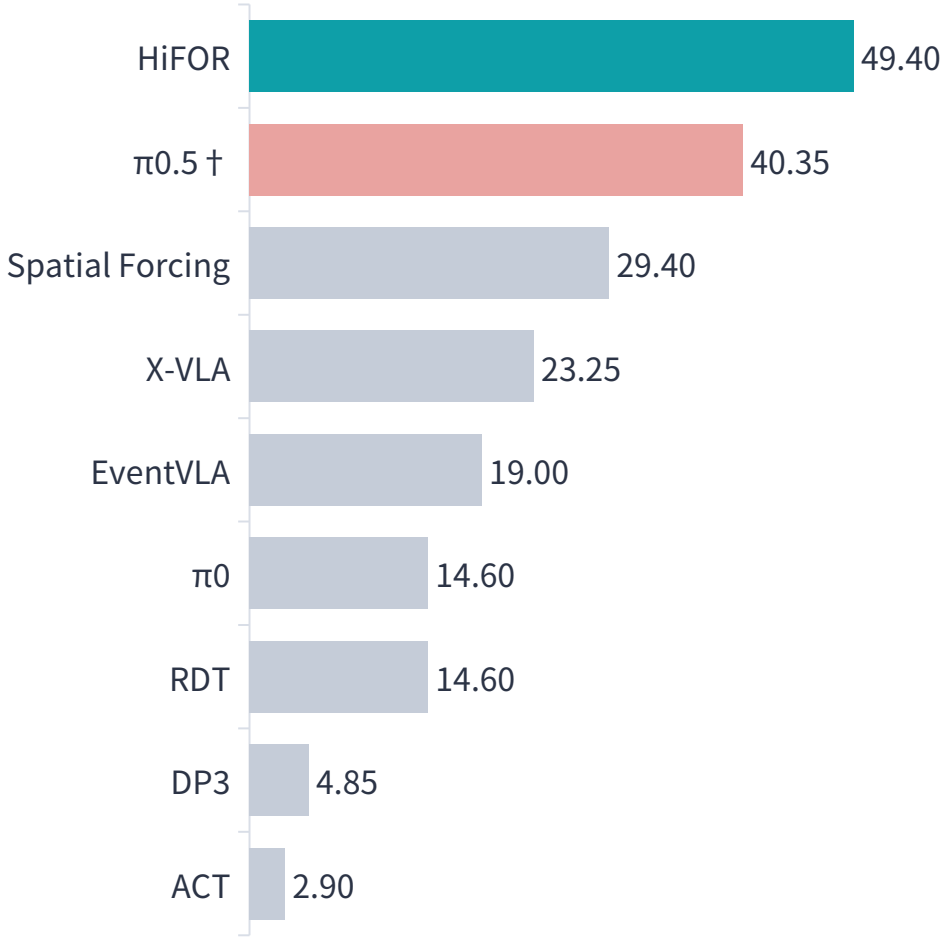
Step 3 Spatial — $\gamma^{p|h,v}$ = Softmax over 16 cells
key: spatial token $k^{h,v,p}$

$$f_{t,\tau} = W_O \cdot \Sigma_{h,v,p} \alpha^h \cdot \beta^{v|h} \cdot \gamma^{p|h,v} \cdot u^{h,v,p}$$

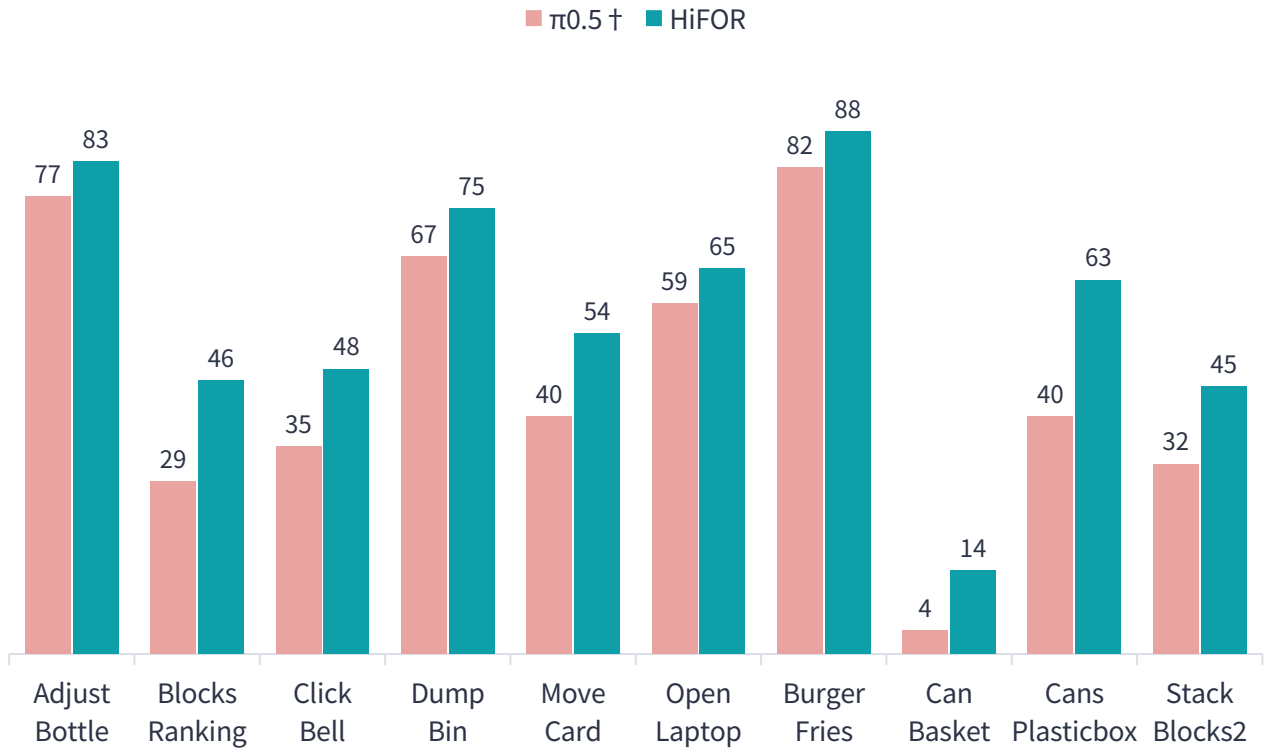
→ Fused with action and short-term state features, residually injected into the Action Expert; the query changes with each denoising step, so retrieval is dynamic too

RoboTwin 2.0 Hard: 49.40% average, 9.05 pts above the strongest baseline π0.5

Average success rate over 20 tasks (%)



10 tasks shown in the main text (%)



All 10 shown tasks improve; largest gains: Cans Plasticbox +23, Blocks Ranking +17, Move Card +14, Stack Blocks +13.

Can Basket stays at only 14%; hard tasks remain unsolved.

LIBERO: tops with 98.3% average; gains come mainly from the long-horizon suite (+3.4)

Method	Spatial	Object	Goal	Long	Avg.
OpenVLA-OFT	97.6	98.4	97.9	94.5	97.1
MemoryVLA	98.4	98.4	96.4	93.4	96.7
π0	96.8	98.8	95.8	85.2	94.2
π0.5 (baseline)	98.8	98.2	98.0	92.4	96.9
VLA-Adapter	97.8	99.2	97.2	95.0	97.3
LEAP-VLA	98.4	**99.4**	97.6	93.0	97.1
HiFOR （Ours）	**99.4**	**99.2**	**98.6**	**95.8**	**98.3**

Excerpt from Table 2; bold cyan = best in column.

LIBERO-Long （vs π0.5）

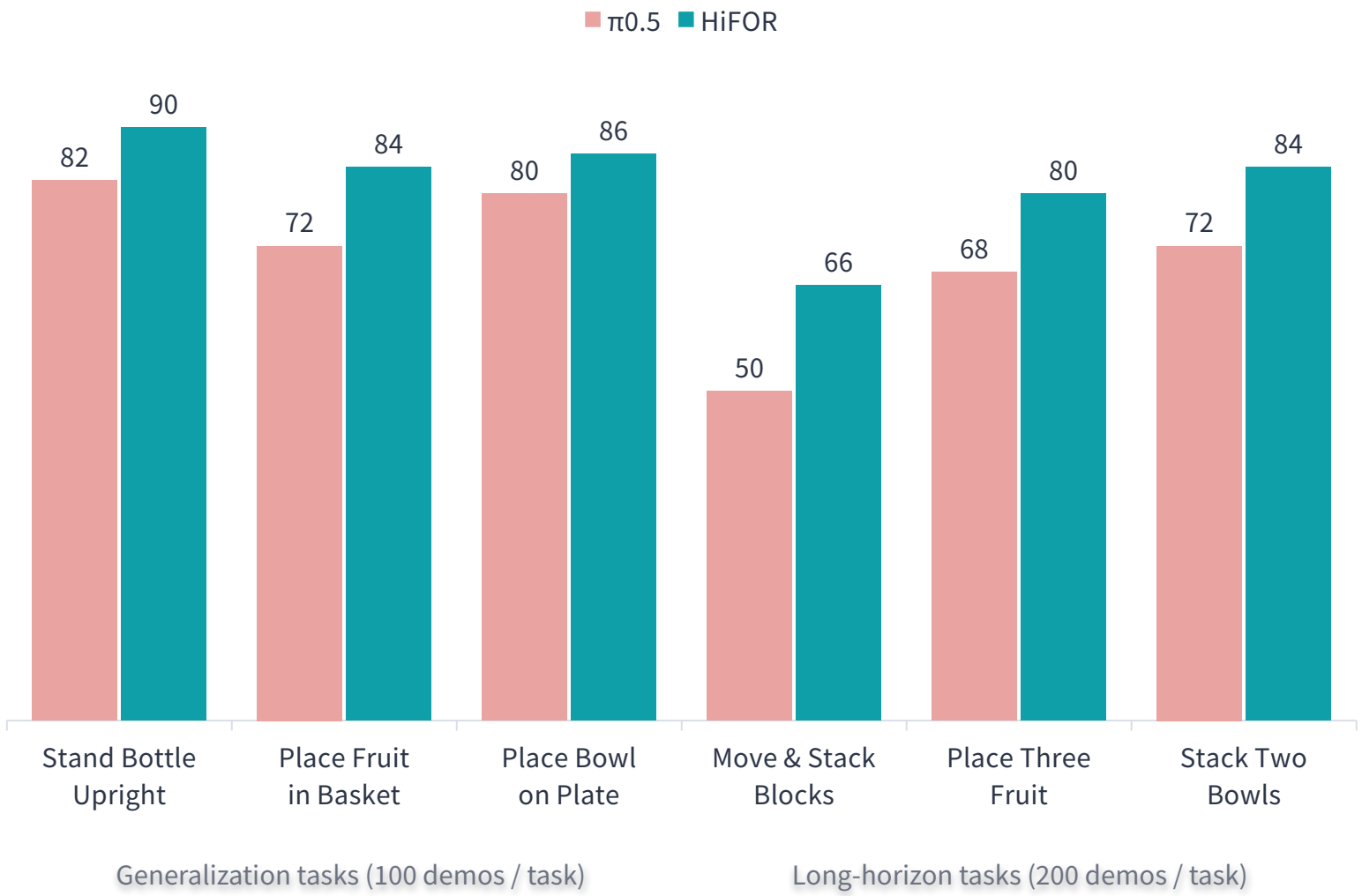
92.4 → **95.8**

+3.4 pts, the largest gain of the four suites

Takeaway

Average 96.9 → 98.3 (+1.4). LIBERO is near saturation, the margin over the runner-up is ~1 pt, and multi-seed variance is unreported; the long-horizon gain is more telling.

Real robot, 6 tasks: all above π0.5, 81.7% average (+11.0), block stacking +16



81.7%

6-task average (π0.5: 70.7%)

+16

Move & Stack Blocks： 50 → 66

Setup

AgileX Piper 6-DoF single arm + parallel gripper

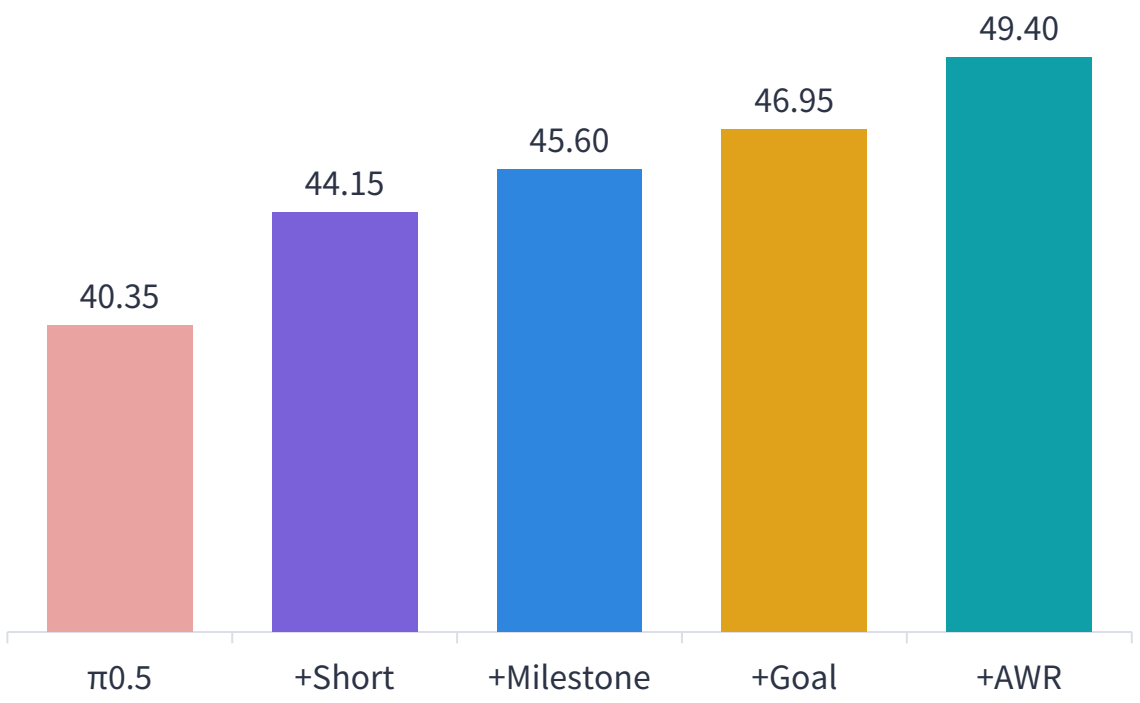
Wrist + third-person: 2× D435i

50 trials per task, randomized object positions

Training 8×H200, deployment 1×RTX 4090

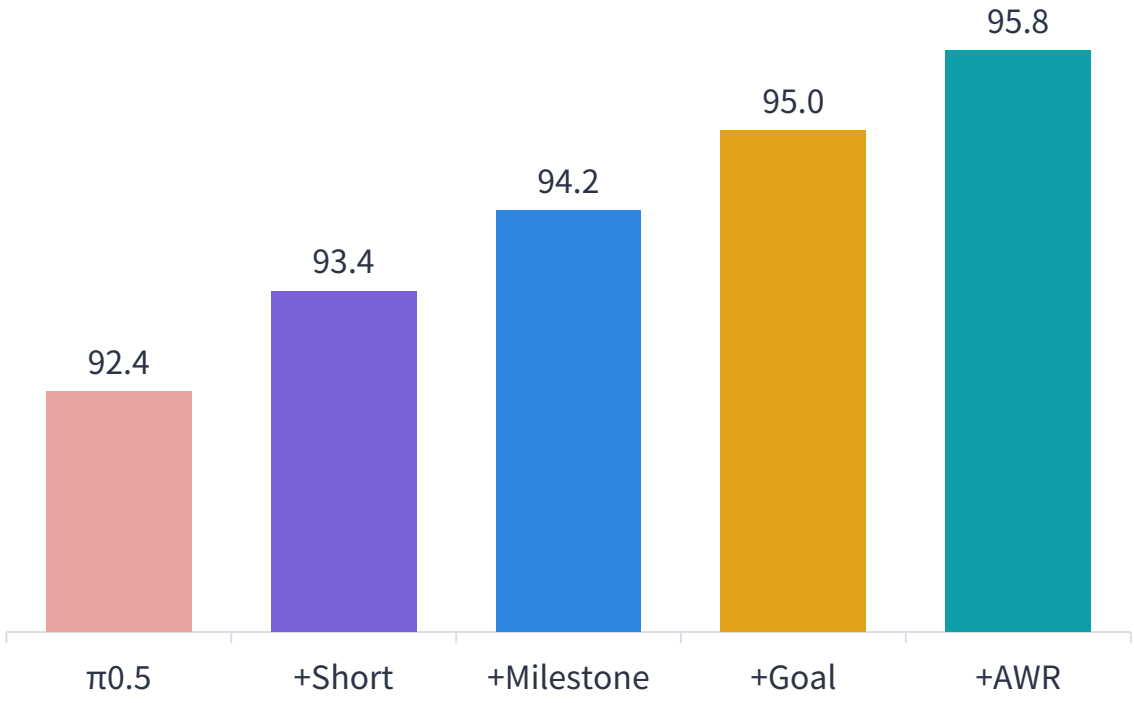
Component ablation: each foresight level helps; short-term prediction and AWR contribute most

RoboTwin 2.0 Hard avg. success rate (%)



LIBERO-Long success rate (%)

Labels below = per-step RoboTwin gains; y-axis not from 0



+3.80 +Short

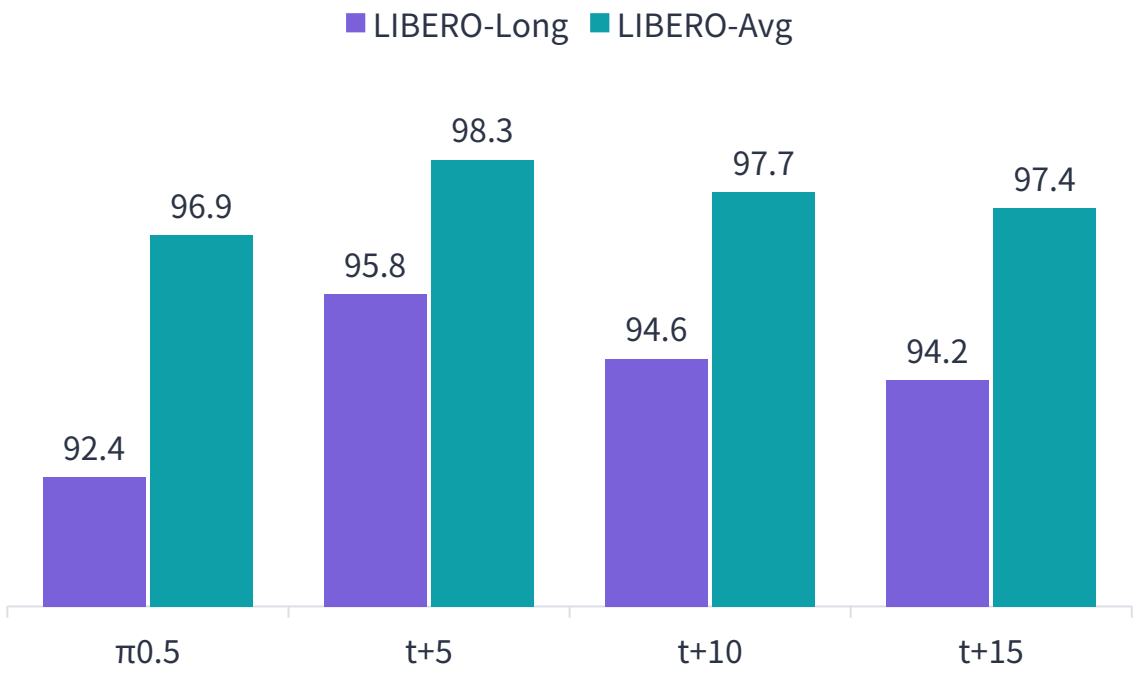
+1.45 +Milestone

+1.35 +Goal

+2.45 +AWR

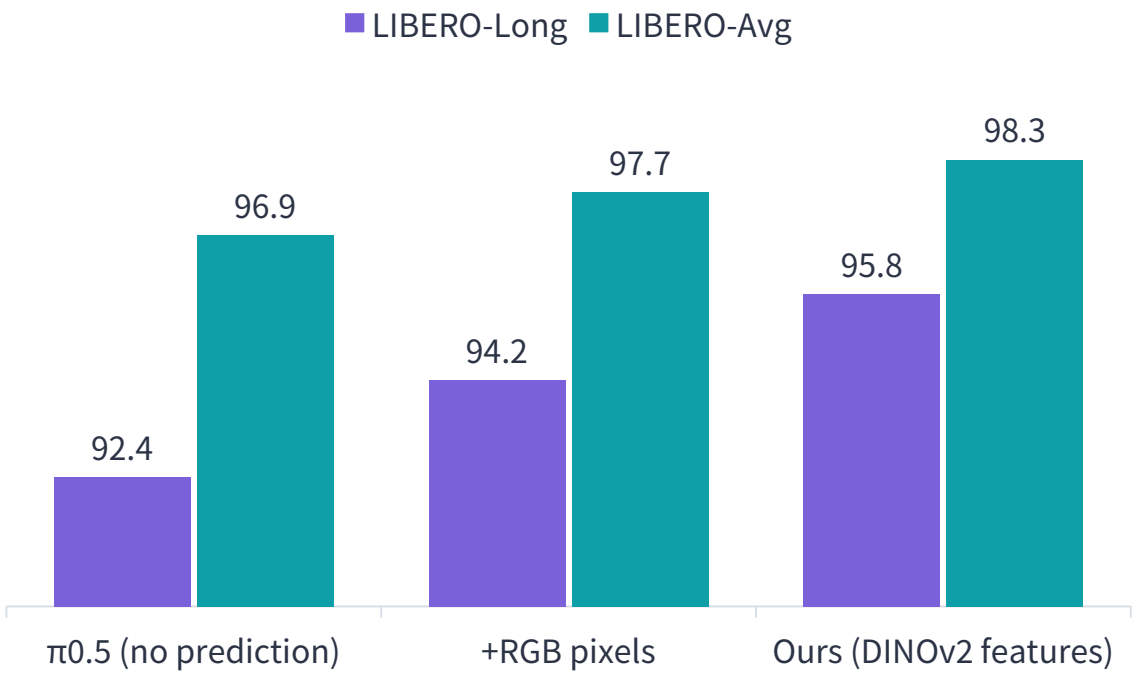
Two design choices: a short-term horizon aligned with the execution cycle works best; predicting semantic features beats pixels

① Short-term prediction length Δ



② Prediction target representation

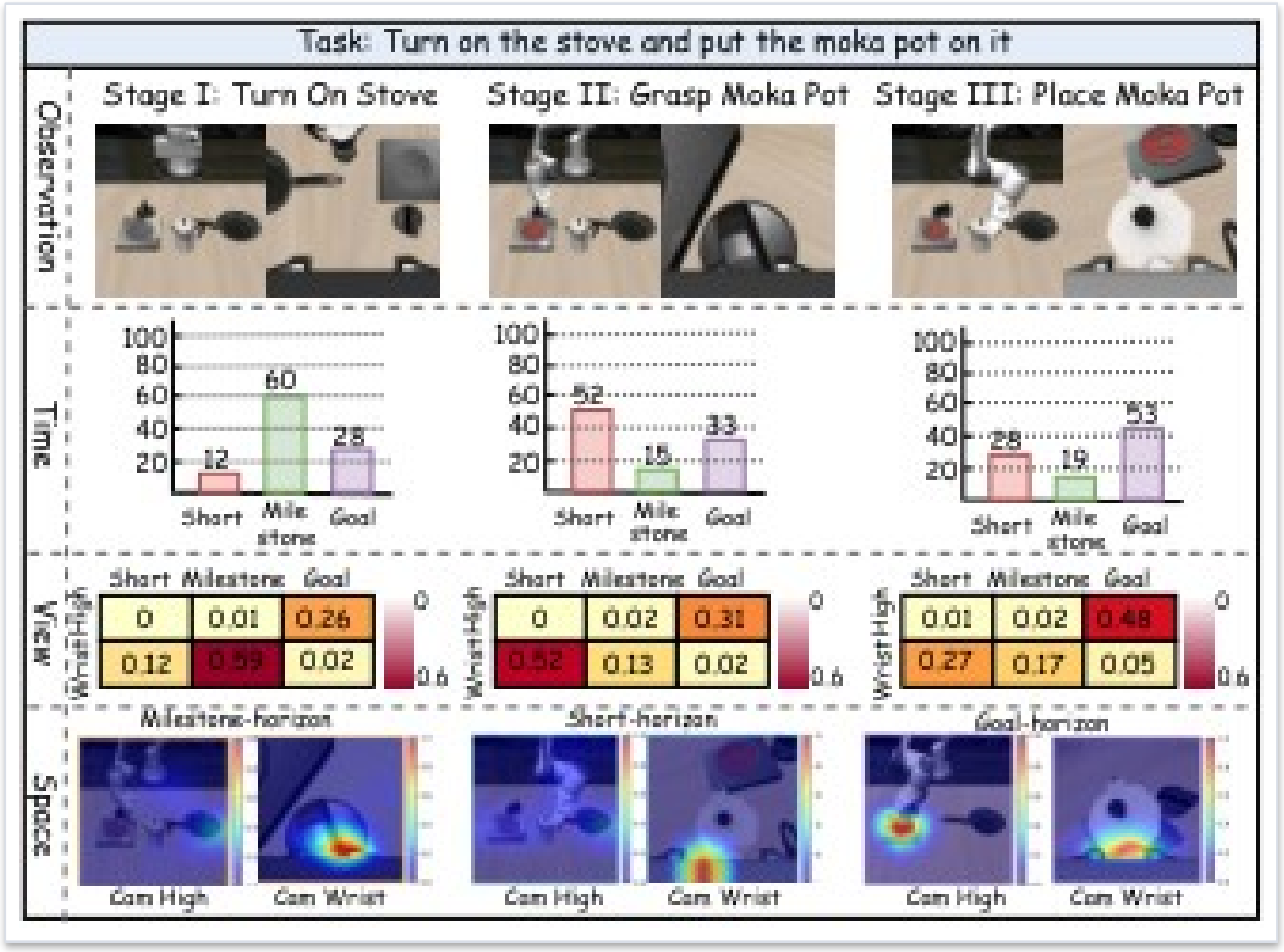
Note: y-axis starts at 90



Δ = 5 is best: LIBERO replans every 5 steps, so this horizon matches the action execution cycle; longer horizons reach past the next update.

Semantic features > pixels: +1.6 on Long, +0.6 avg. vs. RGB prediction; foresight itself helps, and features better guide actions.

Visualization: retrieval attention shifts by stage—milestone to open the stove, short-term to grasp, goal to place



Stage I Open the stove — **60%** milestone



Mostly wrist view, focused on the stove knob

Stage II Grasp the moka pot — **52%** short-term



Wrist view, attending to local manipulation details

Stage III Place the moka pot — **53%** goal



Third-person view, highlighting the placement area

Short-term Milestone Goal (temporal attention %)

Assessment: clear idea and strong long-horizon gains, but weak evidence for key designs' individual contributions

Strengths

Problems ↔ designs map one-to-one

Fragmented time scales → hierarchical foresight + HDM; indiscriminate use → action-guided retrieval

Low supervision cost

Milestones auto-labeled; visual targets are frozen DINOv2 features, no pixel generation needed

Clear long-horizon gains

RoboTwin Hard +9.05; LIBERO-Long +3.4; real-robot stacking +16

Interpretable

Factorized attention directly reveals “which horizon / view / region to attend to”

Limitations & Open Questions

Insufficient ablations

Missing controls: without HDM, flat cross-attention, fixed queries

Statistics & coverage

No multi-seed variance; RoboTwin evaluates only 20 tasks; real robot compared only with π0.5

Annotation generalizability

Relies on gripper-based heuristics + manual review, whose proportion is unreported

Efficiency & details

Inference latency / parameter count unreported; Eq. (6) notation and Table 2 values questionable

Take-aways

01

Hierarchical foresight

Jointly predict short-term dynamics, semantic milestones and final goals, linked by a one-way mask; long-horizon tasks benefit most.

02

On-demand foresight

Organize predictions into a time × view × space structured world, retrieved level by level by the denoising action state instead of plain concatenation.

03

Semantic feature targets

DINOv2 features beat RGB pixels; supervision is built automatically from existing demos at low cost.

In short: HiFOR is a solid incremental step bringing world-model-style foresight into VLAs—consistent gains, yet key designs' individual contributions, efficiency and generalization limits remain unverified.